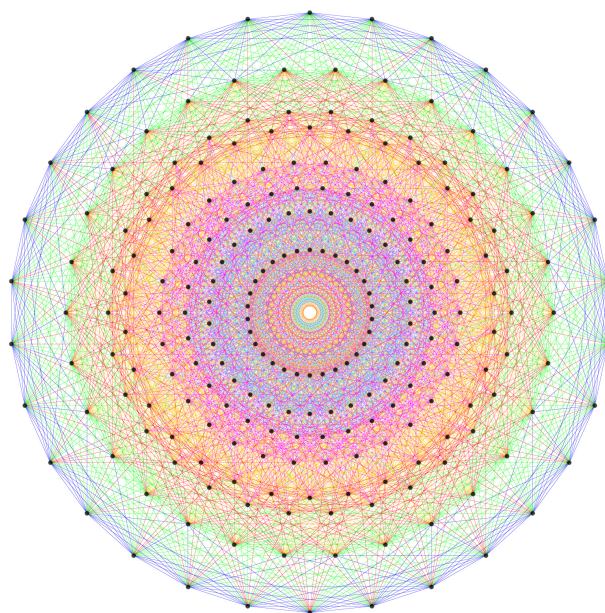


# Representation Theory

Maschke theorem: Brief Exposition

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## Introduction

This exposition aims at exploring the basic foundations of representation theory of finite groups. The main outcome is to explore Maschke's theorem that provides a sufficient condition for when a group algebra is semisimple. We also explore Artin-Wedderburn theorem that characterizes the structure of the semisimple algebras irreducible and finite dimensional representations. The exposition assumes basic knowledge in group, ring and field theory and linear algebra

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# Chapter 1

## Linear Algebra

We begin by defining what is an Algebra. This is the main structure we will be using throughout this paper. To give you an intuition first, an Algebra is basically a set with a structure of both a ring and a vector space.

### 1.1 Algebras over a Field

#### Definition 1.1 ( $k$ -Algebra)

Let  $k$  be any field. We say that a  $k$ -vector space  $A$  is a  $k$ -algebra if  $A$  is equipped with an associative bilinear operation (product)

$$\times : A \times A \rightarrow A$$

satisfying the following axioms of bilinearity. For every  $x, y, z \in A$  and  $a, b \in k$ :

1.  $z \times (x + y) = z \times x + z \times y$
2.  $(x + y) \times z = x \times z + y \times z$
3.  $(a \cdot x) \times (b \cdot y) = ab \cdot (x \times y)$

where  $\cdot$  is the vector space scalar multiplication.

Another way to define a  $k$ -algebra is to first define it as a ring equipped with scalar multiplication operation by embedding the field  $k$  inside  $A$ .

**Definition 1.2** ( $k$ -Algebra)

Let  $k$  be a field. A  $k$ -algebra  $A$  is a ring with an injective ring homomorphism  $k \rightarrow A$ , where for any  $\lambda \in k$  and  $a \in A$  we have that

$$\lambda \cdot a = a \cdot \lambda$$

i.e. multiplication by elements in the image of  $k$  in  $A$  is commutative.

**Remarks:-**

1. Observe that we can define the vector space structure on  $A$  in the definition above by defining the scalar product  $\lambda a$  to be the product of the image of  $\lambda$  in  $A$  times  $a$  to get  $\lambda \cdot a$  for all  $\lambda \in k$  and  $a \in A$ . That is

$$\lambda a = \lambda \cdot a$$

2. An algebra product operation might not be commutative. If so we just say a commutative algebra.
3. We may or may not include 1 as a unity in our definition of an algebra but here we do. And the operation is associative.

**Definition 1.3** ( $k$ -Algebra Homomorphism)

Let  $A, B$  be  $k$ -algebras. A function  $\varphi : A \rightarrow B$  is a  $k$ -algebra homomorphism

if it is a linear map of vector spaces and for any  $a \in A$  and  $b \in B$  we have that

$$\varphi(ab) = \varphi(a)\varphi(b)$$

## 1.2 Direct Sum and Direct Product

Now, we get to differentiate between two important notions in linear algebra: Internal Direct Sum and External Direct Product. Sometimes for simplicity we just say Direct Sum and Direct Product.

### Definition 1.4 (Internal Direct Sum)

Let  $V$  be a vector space. We say  $V$  is an internal direct sum of two subspaces  $U, W \subseteq V$  iff every vector  $v \in V$  could be written uniquely as a sum of two vectors  $u, w$  where  $u \in U$ , and  $w \in W$ .

That is

$$V = U + W.$$

there are unique vectors  $u \in U$  and  $w \in W$  s.t.

$$v = u + w.$$

The above definition tells us exactly when a sum of two subspaces  $U + W$  of a given vector space  $V$  is a direct sum, and in this case we call  $V$  an internal direct sum since the two spaces  $U$  and  $W$  are subspaces. Next, we define an External Direct Product of two separate given spaces.

**Definition 1.5** (External Direct Product)

Let  $K, L$  be two vector spaces. We define the *External Direct Product* of  $K$  and  $L$  to be the vector space

$$K \times L := \{(k, l) \mid k \in K, l \in L\}$$

with operations of component wise addition and scalar multiplication.

The next theorem provides a complete characterization for when an External Direct Product of two subspaces and their sum are exactly the same, giving also an equivalent condition for when a sum is an internal direct sum.

**Theorem 1.6**

Let  $U, W$  be subspaces of a vector space  $V$ . **TFAE**

1. The map

$$\pi : U \times W \rightarrow U + W$$

given by

$$(u, w) \mapsto u + w$$

is an isomorphism.

2.  $U \cap W = \{0\}$ .
3. Any vector  $x \in U + W$  could be written uniquely in the form

$$u + w$$

with  $u \in U$  and  $w \in W$ .

*Proof.* (1)  $\Rightarrow$  (2) Suppose for the sake of contradiction that there exist a non-zero

vector

$$v \in U \cap W.$$

So  $v \in U$  and  $v \in W$ . Hence via the isomorphism  $\pi$ , the element  $(v, 0) \in U \times W$  maps to  $v + 0 = v$  and  $(v, 0) \in U \times W$  maps to  $0 + v = v$  violating injectivity of  $\pi$ .

(2)  $\Rightarrow$  (3) Suppose that there is  $x \in U + W$  such that  $x$  has two different representations

$$x = a + b = c + d.$$

where  $a, c \in U$  and  $b, d \in W$  Then

$$a + b = c + d$$

$$\Rightarrow a - c = d - b.$$

Since  $U, W$  are subspaces that is closed under addition we get that  $a - c \in U$  and  $b - d \in W$ . Since  $a - c = d - b$ , therefore,

$$a - c \in U \cap W.$$

which implies that  $a - c = d - b = 0$ . Hence,  $a = c$  and  $b = d$  as desired.

(3)  $\Rightarrow$  (1) Since any vector  $x \in U + W$  could be written as  $u + w$ , where  $u \in U$  and  $w \in W$  this implies that  $\pi$  is surjective as a preimage for any element  $x = u + w$  is  $(u, w)$ . By uniqueness of this representation of elements in  $U + W$  we get that the preimage of an element  $x = u + w$  is unique. ■

So, if we have a vector space that is a sum of two subspaces

$$U + W = V$$

and satisfies one of the conditions above we say  $V$  is an internal direct sum of  $U, W$

and we write

$$V = U \oplus W.$$

Moreover, we can see that the two notions of internal direct sum and external direct product coincide when  $K, L$  are subspaces of some vector space  $V$  and  $V$  is an external direct sum of  $K$  and  $L$  as proved the direct sum & product are isomorphic.

**Definition 1.7** (Direct sum of Algebras)

Let  $A, B$  be two  $k$ -algebras. We define the direct sum  $A \oplus B$  to be the  $k$ -algebra whose elements are pairs  $(a, b)$  where  $a \in A$  and  $b \in B$ , with component wise operations.

## Chapter 2

# Basics of Representation Theory

### 2.1 Introduction and Motivation

The main goal of representation theory is to understand abstract algebraic structures—specifically groups and associative algebras—by concretely realizing their elements as linear transformations of vector spaces. By transforming non-linear group multiplication into the well-understood domain of linear algebra and matrix arithmetic, one converts difficult group-theoretic questions into tractable eigenvalue problems, trace computations, and spectral decompositions.

First we define what is a representation? then we move to defining a homomorphism and isomorphism of representations. Moreover, we introduce the definition of an important structure that we will be using extensively throughout this exposition which is the group algebra  $k[G]$ , then we will move to working with simple representations/modules. In between we show how can modules and representations be seen as the same thing.

## 2.2 Basic Definitions

### Definition 2.1 (Representation of an Algebra)

Let  $A$  be a  $k$ -algebra, and  $V$  be a  $k$ -vector space. A *representation* of  $A$  is a pair  $(V, \rho)$  where  $\rho : A \rightarrow \text{End}(V)$  is a  $k$ -algebra homomorphism.

### Definition 2.2 (Subrepresentation)

Let  $(V, \rho)$  be a representation of any  $k$ -algebra  $A$ . We say that a subspace  $W \subseteq V$  is a subrepresentation iff elements of  $W$  commute with the representation  $\rho$  or that  $W$  is invariant under the action of  $A$ . That is for any  $a \in A$  and any  $w \in W$  we have that

$$\rho(a) \cdot w \in W$$

### Definition 2.3 (Intertwining Operator)

Let  $(V, \rho_1)$  and  $(W, \rho_2)$  be any two representations of a  $k$ -algebra  $A$ , an intertwining operator  $T : V \rightarrow W$  is a linear map that commutes with the action of  $A$ , that is

$$T(\rho_1(a)(v)) = \rho_2(a)(T(v))$$

We may also say that  $T$  is a homomorphism of representations. If  $T$  is invertible, then it is an isomorphism of representations

### 2.2.1 Equivalence of Representations and Modules

Now, how are representations of any  $k$ -algebra  $A$  could be seen as  $A$ -modules?

We basically define the action of  $A$  on  $V$  by the action of the image of  $A$  in  $\text{End}(V)$  as linear transformations. For the other direction, Let  $M$  be an  $A$ -module. Then  $M$  is an abelian group. We define scalar multiplication via the action of  $A$ , as we mentioned earlier  $A$  has a copy of the field  $k$  inside it, hence: for any  $\lambda \in k$  and

any  $m \in M$  we have that

$$\lambda \cdot m = \lambda m$$

where  $\cdot$  is the action of  $A$  on  $M$ . It remains to prove that the action of  $A$  on  $M$  as a module defines group algebra homomorphism from  $A$  to  $\text{End}(M)$ .

*Proof.* Fix any  $a \in A$ , consider the map

$$\varphi(a) : M \rightarrow M$$

given by  $m \mapsto a \cdot m$ . By the axioms of a module action we get

- $\varphi(a)(m + n) = a \cdot (m + n) = a \cdot m + a \cdot n$
- $\varphi(a)(\lambda m) = a \cdot (\lambda m) = (a\lambda) \cdot m = (\lambda a) \cdot (m) = \lambda(\varphi(a)(m))$

the last equality is because we defined the image of the field  $k$  in  $A$  to be central. So, we proved that  $\varphi(a)$  is a linear map for any  $a \in A$ . Now, we define the group algebra homomorphism

$$\rho : A \rightarrow \text{End}(M)$$

given by  $a \mapsto \varphi(a)$ . To see that it is an algebra homomorphism, first we use the linearity of  $\varphi(a)$

- $\rho(a + b) = \varphi(a + b) = \varphi(a) + \varphi(b) = \rho(a) + \rho(b)$
- $\rho(\lambda a) = \varphi(\lambda a) = \lambda \varphi(a) = \lambda \rho(a)$

since  $M$  is an  $A$  module, then by associativity of the action and the definition of  $\varphi(a)$  for any  $a \in A$ , we get that

- $\rho(ab) = \varphi(ab) = \varphi(a)\varphi(b) = \rho(a)\rho(b)$

■

## 2.3 The Group Algebra

Now, we move to a special type of algebra called the Group algebra  $k[G]$ , that will help us study the representations of a finite group  $G$ .

### Definition 2.4 (The Group Algebra $k[G]$ )

Let  $G$  be a finite group and  $k$  a field. The *group algebra*  $k[G]$  is the  $k$ -vector space with basis given by the elements of  $G$ :

$$k[G] = \left\{ \sum_{g \in G} a_g g \mid a_g \in k \right\}.$$

The multiplication on  $k[G]$  is defined by extending the group multiplication linearly:

$$\left( \sum_{g \in G} a_g g \right) \left( \sum_{h \in G} b_h h \right) = \sum_{g, h \in G} a_g b_h (gh) = \sum_{x \in G} \left( \sum_{gh=x} a_g b_h \right) x.$$

The vector space dimension of  $k[G]$  over  $k$  is precisely the order of the group:

$$\dim_k(k[G]) = |G|.$$

Before we define a representation of a group, we define the general case of a representation of a group algebra.

### Definition 2.5 (Representation of a group algebra $k[G]$ )

Let  $G$  be a group. Let  $k[G]$  be any group algebra over a field  $k$ . A representation of  $k[G]$  is a pair  $(V, \rho)$ , where  $V$  is a  $k$ -vector space and

$$\rho : A \rightarrow \text{End}(V)$$

is a group algebra homomorphism.

The previous definition just says that we are trying to represent elements of an algebra with linear transformation or matrices.

## 2.4 Representation of a Group

We are done discussing the general case. But what about the definition of a group representation? There are two ways to view this definition. One way is to see it as a restriction of the definition of an algebra representation on the basis elements which are basically the elements of the group  $G$ . The other one, which I find more interesting is to see it as an extension of a group action.

First, recall the definition of a group action on an arbitrary set  $S$ .

### Definition 2.6 (Group Action)

Let  $G$  be a multiplicative group, and  $S$  to be any set. We define a group action of  $G$  on  $S$  to be a function

$$\cdot : G \times S \rightarrow S$$

denoted by  $g \cdot x$  for all  $g \in G$  and  $x \in S$ , satisfying the following axioms:

1.  $e \cdot x = x \quad \forall x \in S$
2.  $g \cdot (h \cdot x) = gh \cdot x$

Using the previous two axioms of a group action, one can see that it actually induces a group homomorphism  $\alpha : G \rightarrow \text{Sym}(S)$ , where  $\text{Sym}(S)$  is the group of all permutations on  $S$  with function composition.

*Proof.* Let  $\cdot : G \times S \rightarrow S$  be a group action. Consider the map

$$\alpha : G \rightarrow \text{Sym}(S)$$

where  $\alpha(g)$  is a bijection on  $S$  such that

$$\alpha(g)(x) = g \cdot x$$

for all  $g \in G$  and  $x \in S$ . We claim that  $\alpha$  is a group homomorphism. By the axioms of the group action, for any  $g, h \in G$ :

$$\alpha(gh)(x) = gh \cdot x = g \cdot (h \cdot x) = \alpha(g)(\alpha(h)(x)) = \alpha(g) \circ \alpha(h)(x)$$

■

Now, we repeat the same process but using a group action of a group  $G$  on a vector space  $V$  instead of an arbitrary set  $S$  requiring an additional property for the group action which is to be *linear*.

**Definition 2.7** (Linear Group Action)

Let  $V$  be a  $k$ -vector space, and  $G$  be any group. Define the group action of  $G$  on  $V$  as the function

$$\cdot : G \times V \rightarrow V$$

that satisfies the following axioms:

1.  $e \cdot v = v \quad \forall v \in V$
2.  $g \cdot (h \cdot v) = gh \cdot v$
3.  $g \cdot (v + w) = g \cdot v + g \cdot w$
4.  $g \cdot (\lambda v) = \lambda(g \cdot v)$

This definition is in fact equivalent to having a group homomorphism  $\varphi : G \rightarrow \text{GL}(V)$ , exactly in the same way a regular group action gives rise to a homomorphism of groups from  $G$  to the symmetric group on an arbitrary set  $S$ . This pair of the homomorphism  $\varphi$  along with the vector space  $V$  are called a *representation* of the group  $G$ .

**Definition 2.8** (Representation of a Group)

Let  $G$  be a group, and let  $V$  be a vector space over a field  $k$ . A *representation* of  $G$  is a pair  $(V, \varphi)$  where  $\varphi : G \rightarrow \text{GL}(V)$  is a group homomorphism given by the linear action of  $G$  on  $V$ .

**Definition 2.9** ( $G$ -Homomorphism)

Let  $(\rho_1, V_1)$  and  $(\rho_2, V_2)$  be two representations of a group  $G$  over  $k$ . A linear map  $T : V_1 \rightarrow V_2$  is called an *intertwining operator* (or  *$G$ -homomorphism*) if it commutes with the  $G$ -action:

$$T \circ \rho_1(g) = \rho_2(g) \circ T \quad \text{for all } g \in G.$$

If  $T$  is an invertible linear isomorphism, we say that  $\rho_1$  and  $\rho_2$  are *equivalent* (or *isomorphic*), denoted  $\rho_1 \cong \rho_2$ .

**Definition 2.10** (Subrepresentation)

Let  $\rho : G \rightarrow \text{GL}(V)$  be a representation. A vector subspace  $W \subseteq V$  is called a  *$G$ -invariant subspace* (or *subrepresentation*) if

$$\rho(g)w \in W \quad \text{for all } g \in G \text{ and all } w \in W.$$

**Definition 2.11** (Irreducible representation)

We say a representation  $(V, \rho)$  is irreducible if  $V$  has no proper non-zero subrepresentations.

**Definition 2.12** (Indecomposable representation)

We say a representation  $(V, \rho)$  is indecomposable if there doesn't exist a pair of subrepresentations  $U, W$  such that  $V = U \oplus W$ .

**Note.** It is essential to distinguish between *irreducibility* and *indecomposability*:

- A representation  $V$  is *irreducible* if it admits no proper non-zero subrepresentation.
- A representation  $V$  is *indecomposable* if it cannot be written as a direct sum of two non-zero subrepresentations, i.e.,  $V \neq W_1 \oplus W_2$  with  $W_1, W_2 \neq \{0\}$ .

Every irreducible representation is indecomposable, but the converse does not hold in general fields! As we shall see in Section ??, when the characteristic of the field divides the group order, there exist indecomposable representations that are strictly reducible.

## Chapter 3

# Semi-Simple Algebras

### 3.1 Definitions

Now, as we are done talking about the basics of representation theory, we get to the main goal of this exposition: introducing Semisimple algebras. The main claim we aim to prove in this chapter is that all semisimple algebras are product of matrix algebras. This characterization shows that semisimple algebras are the easiest to work with.

**Definition 3.1** (Semisimple Algebra)

An algebra  $A$ , viewed as a representation of  $A$  is *semisimple* or *completely reducible* if it is a direct sum of irreducible representations of  $A$ .

There are many ways to define semisimple algebras, and they are all equivalent. At the end, we see that all of these definitions are just stating properties that such an algebra has. We will prove that all of these properties are equivalent.

In fact, semisimple algebras are nice to work with since they are all isomorphic to a direct sum of matrix algebra (Artin-Wedderburn). Moreover, we will see a characterization for an arbitrary group algebra over a field (with some restrictions on the characteristic of that field) to be semisimple thus nice, and this is the main

title of this exposition which is Maschke's Theorem.

Now, we have a nice lemma about irreducible representations that makes a lot of sense if we can see that the kernel and the image of an intertwining operator are subrepresentations.

**Lemma 3.2**

Let  $A$  be a  $k$ -algebra and  $V, W$  be two irreducible representations of  $A$ . Let  $\varphi : V \rightarrow W$  be an intertwining operator. Then  $\varphi$  is either zero or an isomorphism.

*Proof.* Since  $\ker \varphi$  is a subrepresentation of  $V$  then  $\ker V = 0$  or  $\ker V = V$ .

1. Case 1: If  $\ker V = 0$ , then  $\varphi$  is injective. But also  $W$  is irreducible so  $\text{Img } \varphi = 0$  or  $\text{Img } \varphi = W$  by rank-nullity we get that  $\text{Img } \varphi$  must be  $W$  hence  $\varphi$  is an isomorphism.
2. Case 2: if  $\ker V = V$  then  $\varphi$  is trivially the zero map.

and this ends the proof. ■

before we go to the main theorems we define the regular and trivial representations

**Definition 3.3** (Regular Representations)

**Definition 3.4** (Trivial Representation)

**Definition 3.5** (Radical)

**Theorem 3.6** (Finitely many irreducible representations)

**Theorem 3.7** (The irreducible representations of a semisimple algebra)

**Theorem 3.8** (Equivalences to semisimplicity)

**Theorem 3.9** (Density Theorem)

**Theorem 3.10** (Maschke Theorem)

## Chapter 4

# Conclusion

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## Appendix A

# Appendix Title

Appendix goes here...

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